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Switch circuit for implement a soft start

Abstract

A switch circuit includes: a p-channel pass transistor, in which the source is connected to a power source and the drain is connected to a load; and a gate drive circuit, which is connected to the gate of the pass transistor. The gate drive circuit includes: a first transistor, through which a first current is made to flow by turn-on of an on/off signal; and an amplifier, in which a reference voltage corresponding to the first current is input to the negative input end, and a gate voltage of the gate is input to the positive input end, and which controls a gate current from the gate by an output from the output end in a manner that the gate voltage matches the reference voltage, and is capable of setting a maximum current of the output. A soft-on time from turn-on of the on/off signal to turn-on of the pass transistor is set by setting the maximum current according to the gate capacitance of the pass transistor.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The disclosure relates to a switch circuit using a p-channel pass transistor disposed between a power source and a load.

2. Description of the Related Art

(2) Conventionally, as a switch circuit for controlling a current from a power source to a load, a

switch circuit using a pass transistor which is a metal-oxide-semiconductor field-effect transistor (MOSFET) is known.

(3) Here, when an n-channel pass transistor is disposed at the power source side of the load, a voltage higher than the power source voltage is required to turn the pass transistor on and off. Thus, a charging pump or the like is required, which makes the configuration complicated.

(4) Therefore, when the pass transistor is disposed at the power source side, it is preferable to use a p-channel MOSFET.

(5) Moreover, when the pass transistor is turned on suddenly, an inrush current to the load is generated, and thus the pass transistor needs to be turned on gradually (soft-on), thus a circuit for the suitable soft-on is required.

SUMMARY OF THE INVENTION

(6) A switch circuit related to the disclosure includes: a p-channel pass transistor, in which the source is connected to a power source and the drain is connected to a load; and a gate drive circuit, which is connected to the gate of the pass transistor, wherein the gate drive circuit includes: a first transistor, through which a first current is made to flow by turn-on of an on/off signal; and an amplifier, in which a reference voltage corresponding to the first current is input to the negative input end, and a gate voltage of the gate is input to the positive input end, and which controls a gate current from the gate by an output from the output end in a manner that the gate voltage matches the reference voltage, and is capable of setting a maximum current of the output, and a soft-on time from turn-on of the on/off signal to turn-on of the pass transistor is set by setting the maximum current according to the gate capacitance of the pass transistor.

(7) According to the switch circuit related to the disclosure, the generation of an inrush current can be prevented by using the gate capacitance of a pass transistor, thereby implementing a soft start.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a circuit diagram showing an overall configuration of a switch circuit according to an embodiment of the present invention.

(2) FIG. 2 is a timing chart showing the state of each part when $V_{in} > I_1 \cdot R_1$.

(3) FIG. 3 is a timing chart showing the state of each part when $V_{in} < I_1 \cdot R_1$.

PREFERRED EMBODIMENT OF THE PRESENT INVENTION

(4) Hereinafter, embodiments of the disclosure are described with reference to the drawings. Note that, the following embodiments do not limit the scope of the disclosure, and configurations obtained by selectively combining multiple examples are also included in the disclosure.

(5) “Overall Configuration”

(6) FIG. 1 is a circuit diagram showing an overall configuration of a switch circuit according to an embodiment of the present invention. A pass transistor M12 is turned on or off according to an on/off signal ON/OFF so as to control power supply to a load RL.

(7) The source of the p-channel pass transistor M12 is connected to a power source V_{in} (power source voltage= V_{in}), one end of the load R1 is connected to the drain of the pass transistor M12, and the other end of the load RL is connected to a ground gnd. Therefore, the current from the power source V_{in} flows through the load RL by turning on the pass transistor M12. Note that, the pass transistor M12 is a p-channel MOSFET, and the drain is the output end Vout.

(8) In the embodiment of the present invention, a gate drive circuit 10 and a monitor circuit 20 are provided as circuits for controlling the pass transistor M12.

(9) <Gate Drive Circuit 10>

(10) First, the gate drive circuit 10 will be described as following. One end of a resistor R1 is connected to the power source V_{in} , and the drain of an n-channel transistor M1 serving as a first

transistor, is connected to the other end of the resistor **R1** via a current source **CS1**. In addition, the source of the transistor **M1** is connected to the ground **gnd**. A constant current **i1** is made to flow by the current source **CS1**. Therefore, when the transistor **M1** is turned on, the current **i1** flows through the resistor **R1** and the transistor **M1**. Note that, the current **i1** is referred to as the first current.

(11) The on/off signal **ON/OFF** is supplied to the gate of the transistor **M1**, and the gate of the transistor **M1** is turned on or off according to the on/off signal **ON/OFF**.

(12) The source of a p-channel transistor **M3** is connected to the power source **Vin**, and the drain of an n-channel transistor **M2** serving as a second transistor is connected to the drain of the transistor **M3** via a current source **CS2**. In addition, the source of the transistor **M2** is connected to the ground **gnd**.

(13) A current **i2** serving as the second current is made to flow by the current source **CS2**. Therefore, if the transistor **M2** is turned on, the current **i2** flows through the transistor **M3** and also the transistor **M2**. The on/off signal **ON/OFF** is supplied to the gate of the transistor **M2** via an inverter **inv1**, and the gate of the transistor **M2** is turned on or off according to an inverted signal of the on/off signal **ON/OFF**.

(14) Therefore, when the on/off signal **ON/OFF** is **ON** (at an H level), the transistor **M1** is turned on and the transistor **M2** is turned off, and when the on/off signal **ON/OFF** is **OFF** (at an L level), the transistor **M1** is turned off and the transistor **M2** is turned on.

(15) The transistor **M3** is short-circuited between the gate and the drain, and the gate of the transistor **M3** is connected to the gate of a p-channel transistor **M4** having the source connected to the power source **Vin**. Therefore, the transistor **M3** and the transistor **M4** constitute a current mirror. Here, the area ratio of the transistor **M3** to the transistor **M4** is 1:m, and m times the current **i2** of the transistor **M3**, that is, a current $m \cdot i2$, flows through the transistor **M4**.

(16) The drain of the transistor **M4** is input to the positive input end **ip** of a gm amplifier **A1**, and is connected to the gate of the pass transistor **M12**. A voltage **vref** of the connection point between the resistor **R1** and the current source **CS1** is input to the negative input end **in** of the gm amplifier **A1**. Therefore, the gm amplifier **A1** operates in a manner that a voltage of the positive input end **ip**, that is, the gate voltage **Vg** of the pass transistor **M12**, becomes the voltage **vref**. Note that, the voltage **vref** is referred to as the reference voltage.

(17) The drain of an n-channel transistor **M5** is connected to the output end of the gm amplifier **A1**, and the source of the transistor **M5** is connected to the ground **gnd**. The transistor **M5** is short-circuited between the gate and the drain, and the gate of the transistor **M5** is connected to the gate of an n-channel transistor **M6** having the source connected to the ground **gnd**. Therefore, the transistor **M5** and the transistor **M6** constitute a current mirror. The drain of the transistor **M6** is connected to the positive input end of the gm amplifier **A1** and the gate of the pass transistor **M12**. Furthermore, a connection point of the drain of the transistor **M6** with the positive input end of the gm amplifier **A1** and the gate of the pass transistor **M12** is connected to the power source **Vin** via a resistor **R2**. Therefore, the gate of the pass transistor **M12** is pulled up to **Vin** when the gate current is not controlled. Furthermore, the area ratio of the transistor **M5** to the transistor **M6** is 1:n, and a current which is n times the current flowing through the transistor **M5** flows through the transistor **M6**.

(18) Here, the output current of the gm amplifier **A1** flows through the transistor **M5**, and a current which is n times the output current of the gm amplifier **A1** flows through the transistor **M6**, so that electric charges are extracted from the gate of the pass transistor **M12**, and accordingly the voltage of the positive input end **ip** decreases. Therefore, when the voltage of the positive input end **ip** is higher than **vref**, the output current increases, and a maximum value of the output current (=saturation current) of the gm amplifier **A1** is defined as the current **imax**. Note that, the current **imax** is referred to as the maximum current. In addition, in this example, the voltage **vref** is set according to the gate voltage **Vg** at which the pass transistor **M12** is fully turned on.

(19) Therefore, when the on/off signal ON/OFF is ON, the current i_1 flows, and the gate voltage V_g of the pass transistor M12 is sufficiently higher than the voltage v_{ref} , the gm amplifier A1 outputs the current i_{max} . Therefore, the gate current $i_g = n \cdot i_{max}$ is extracted from the gate of the pass transistor M12 via the transistor M6.

(20) On the other hand, if the on/off signal ON/OFF is OFF, the current i_2 flows through the transistor M3, that is, if the gate voltage V_g of the pass transistor M12 is lower than the power source voltage V_{in} , the current i_2 will flow through the transistor M3. Therefore, the gate current $i_g = m \cdot i_2$ is supplied to the gate of the pass transistor M12 via the transistor M4.

(21) <Monitor Circuit 20>

(22) Next, the monitor circuit 20 will be described. First, a first monitor circuit will be described as following. The gate of an n-channel transistor M7 is commonly connected to the gates of the transistors M5 and M6 that constitute the current mirror, and the source of the transistor M7 is connected to the ground gnd . The area of the transistor M7 is the same as that of the transistor M5, and the current i_{max} is made to flow by the transistor M7.

(23) The drain of the transistor M7 is connected to the power source V_{in} via a current source CS3. And, the voltage of a connection point between the current source CS3 and the drain of the transistor M7 is output as a signal g_charge via an inverter inv2. A current i_3 is set to be smaller than the current i_{max} ($i_{max} > i_3$), when the current i_{max} flows, the signal g_charge , which is the output of the inverter inv2 becomes the L level, and when no current i_{max} flows, the signal g_charge becomes the H level.

(24) A second monitor circuit will be described as following. The drain of an n-channel transistor M8 is connected to the gate of the pass transistor M12. The gate of an n-channel transistor M9 is connected to the gate of the transistor M8, and these gates are set to a voltage v_b . The voltage v_b is a voltage slightly higher than the threshold voltages of the transistors M8 and M9. In addition, the sources of the transistors M8 and M9 are connected to each other. And, the drain of the transistor M9 is connected to the power source V_{in} via a current source CS4, and the voltage of the connection point of the drain of the transistor M9 with the current source CS4 is output as a signal $vg0v$ via an inverter inv3.

(25) As described above, the voltage v_b is set to a voltage slightly higher than the threshold voltages of the transistors M8 and M9. Therefore, generally, no current flows through the transistors M8 and M9, and the signal $vg0v$ is at the L level.

(26) On the other hand, when the on/off signal ON/OFF is ON and $V_{in} < i_1 \cdot R_1$, the gate voltage V_g becomes nearly a gnd level (0 V in this example) trying to allow the transistor M12 to be fully turned on. In this case, when the current flows via the transistors M9 and M8 and the current amount of the current is greater than that of a current i_4 , the signal $vg0v$ becomes the H level.

(27) A third monitor circuit will be described as following. The drain of a p-channel transistor M10 having the source connected to the power source V_{in} is connected to the ground gnd via a current source CS5. The voltage of a connection point of the drain of the transistor M10 with the current source CS5 is output as a signal $gate_off$ via an inverter inv4. The gate voltage V_g of the pass transistor M12 is lower than that of the power source V_{in} by a predetermined value, and when the pass transistor M12 is turned on, the current flows through the transistors M11 and M10, and the signal $gate_off$ becomes the L level. That is, by setting a resistor R3 to a value defined by the ratio of a potential difference between the threshold voltage of the gate of the transistor M12 and the threshold voltage of the transistor M11 to a current i_5 , the transistor M12 is turned off below a desired threshold voltage, and accordingly the signal $gate_off$ becomes the L level.

(28) "Drive Operation"

(29) FIG. 2 is a timing chart showing the state of each part when $V_{in} > i_1 \cdot R_1$. First, if the on/off signal ON/OFF is OFF (at the L level), the transistor M1 is turned off and the transistor M2 is turned on. In this case, the voltage $v_{ref} = V_{in}$, $V_g = V_{in}$, and the gate current $i_g = 0$. In addition, when the current of the transistor M7 is 0, the signal g_charge is at the L level, when the current of the

transistor M9 is 0, the signal vg_{0v} is at the L level, and when the current of the transistor M10 is 0, the signal $gate_off$ is at the H level.

(30) In this state, when the on/off signal ON/OFF is ON (at the H level), the transistor M1 is turned on, the current i_1 flows, and the voltage $v_{ref}=V_{in}-i_1*R_1$. Then, the voltage v_{ref} is input to the negative input end of the gm amplifier A1. The voltage of the positive input end of the gm amplifier A1 is V_{in} , and the current i_{max} flows in the output of the gm amplifier A1. Therefore, the current of $i_{max}*n$ flows through the transistor M6, and is extracted from the gate of the pass transistor M12.

(31) Here, the gate of the pass transistor M12 has a gate capacitance. The pass transistor M12 is a transistor that supplies a current to the load R_L and has a relatively large gate capacitance C_g . At a stage that the on/off signal ON/OFF is OFF, the gate voltage is V_{in} , and the electric charge charged to the gate is C_g*V_{in} . Then, i_{max} is output from the gm amplifier A1, and the current $i_{max}*n$ flows through the transistor M6, and is extracted from the gate of the pass transistor M12

($i_g=i_{max}*n$). Then, the gate voltage V_g decreases from V_{in} to v_{ref} . Here, $V_{in}-v_{ref}=i_1*R_1$, and the soft-on time t_{on} required for the gate voltage to change from V_{in} to v_{ref} can be expressed as follows.

$$t_{on}=C_g*i_1*R_1/(i_{max}*n)$$

(32) Note that, the maximum value of the gate-to-source voltage V_{gs} of the pass transistor M12 is i_1*R_1 . In addition, capacitors may be separately connected in parallel between the gate and the source to adjust the capacitance between the gate and the source.

(33) And, in a state that the gate voltage $V_g=v_{ref}=V_{in}-i_1*R_1$, the pass transistor M12 is fully turned on and supplies a current to the load R_L .

(34) Here, if $V_{gs}<i_1*R_1$, the output current of the gm amplifier A1 is i_{max} , and the current of the transistor M7 is greater than the current 3. Therefore, the gate current $i_g=i_{max}*n$ flows through the transistor M6, and accordingly the signal g_charge becomes the H level. Furthermore, the output current of the gm amplifier A1 becomes 0 at a stage that $V_g=v_{ref}$, and thus the signal g_charge becomes the L level. That is, the gate current is extracted from the gate of the pass transistor M12, and the pass transistor M12 becomes the H level in the time from being turned off to being turned on, that is, in the soft-on time.

(35) The condition of this example is that a voltage decrease i_1*R_1 at the resistor R_1 is less than the power source V_{in} ($V_{in}>i_1*R_1$). Therefore, the currents of the transistors M9 and M8 are never greater than the current i_4 , and the signal vg_{0v} remains at the L level.

(36) In addition, when the on/off signal ON/OFF becomes on, the gate current i_g of the pass transistor M12 flows, and the gate voltage V_g starts to decrease, the current starts to flow through the transistor M11. The current is relatively small due to the resistor R_3 , and can be negligible for the gate current i_g . Note that, the current flowing through the transistor M11 need not necessarily be set to a very small current, and in this case, the current amount in relation to the current i_{max} may be taken into consideration. And, when the current flows through the transistor M11, the current flows through the transistor M10, and the current amount of the current is set to be greater than that of the current 5, and accordingly the signal $gate_off$ becomes the L level.

(37) Next, when the on/off signal ON/OFF becomes OFF, the transistor M1 is turned off, the current i_2 flows through the transistors M2 and M3, and the current $m*i_2$ flows through the transistor M4. Then, this current $m*i_2$ is supplied to the gate of the pass transistor M12 ($i_g=m*i_2$), and the gate voltage V_g of the pass transistor M12 starts to rise. That is, the gate capacitance C_g is discharged until the gate voltage reaches V_{in} . Therefore, a soft-off time t_{off} required for the gate voltage to change from v_{ref} to V_{in} can be expressed as follows.

$$t_{off}=C_g*i_1*R_1/(i_2*m)$$

(38) Moreover, the gate voltage of the pass transistor M12 rises with the turn-off of the on/off signal ON/OFF, and accordingly the current of the transistor M11 decreases, and the signal $gate_off$ becomes the H level. Note that, although the area ratio of the transistors M10 and M11 constituting

a current mirror is set to 1:1 in this example, the disclosure is not limited thereto.

(39) FIG. 3 is a timing chart showing the state of each part when $V_{in} < i_1 \cdot R_1$. In response to the turn-on of the on/off signal, the voltage v_{ref} and the gate voltage V_g of the pass transistor M12 intends to become $V_{in} - i_1 \cdot R_1$, but become nearly 0 V. In addition, in response to the turn-off of the on/off signal, the voltage v_{ref} and the gate voltage V_g of the pass transistor M12 return to V_{in} . The behaviors of the gate current i_g and the signals g_charge and $gate_off$ are also the same as that in FIG. 2.

(40) Here, the current i_{max} flows through the transistor M7 by the turn-off of the on/off signal ON/OFF, and accordingly the signal g_charge becomes the H level. And, the current i_{max} becomes 0 by fully turning on the pass transistor M12. However, when V_{in} is sufficiently lower than $i_1 \cdot R_1$ ($V_{in} < i_1 \cdot R_1$), the transistor M12 still does not reach the full-on state even if the gate voltage V_g decreases to near 0 V. As a result, the g_charge indicating the state of being charged remains H. The reason is that the amplifier A1 compares v_{ref} which is the target with V_g , and when V_{in} is close to $i_1 \cdot R_1$, the determination of g_charge becomes uncertain due to the limit of the operation region near the gnd level (0 V in this example) of each element (M1, CS1, A1, M6). In FIG. 3, this uncertain state is indicated by a plurality of diagonal lines.

(41) For example, the gate voltage V_g of the transistor M12 is constant at the gnd level, and thus no current flows through the transistor M12, but instead, the current flows towards the transistor M6. The reason is that the amplifier A1 defines that the output current is insufficient and outputs almost the current i_{max} . Then, the current i_{max} is output, and accordingly the current flows through the transistors M9 and M8, and the signal vg_{0v} becomes the H level.

(42) When the on/off signal ON/OFF becomes OFF, the transistor M6 is turned off, and the currents of the transistors M8 and M9 have nowhere to flow, then the drain voltage of the transistor M9 rises, and the signal vg_{0v} returns to the L level.

(43) The embodiment of the present invention includes the signal vg_{0v} . Therefore, by gating the signal g_charge with an AND gate in which the signal vg_{0v} is input to the other end, an erroneous output of the H level in the signal g_charge can be prevented.

Claims

1. A switch circuit, comprising: a p-channel pass transistor, in which a source is connected to a power source and a drain is connected to a load; and a gate drive circuit, which is connected to a gate of the pass transistor, wherein the gate drive circuit comprises: a first transistor, through which a first current is made to flow by turn-on of an on/off signal; and an amplifier, in which a reference voltage corresponding to the first current is input to a negative input end, and a gate voltage of the gate of the pass transistor is input to a positive input end, and which controls a gate current from the gate of the pass transistor by an output from the output end of the amplifier in a manner that the gate voltage of the gate of the pass transistor matches the reference voltage, while the gate voltage is higher than the reference voltage, an output current of the amplifier increases, a maximum value of the output current is defined as a maximum current of the output of the amplifier, and a soft-on time from turn-on of the on/off signal to turn-on of the pass transistor is set by setting the maximum current according to a gate capacitance of the pass transistor.

2. The switch circuit according to claim 1, further comprising: a second transistor, through which a second current is made to flow by turn-off of the on/off signal, wherein a soft-off time from turn-off of the on/off signal to turn-off of the pass transistor is set by passing a current corresponding to the second current as a gate current to the gate.

3. The switch circuit according to claim 1, further comprising: a first monitor circuit, which comprises a current mirror passing a current corresponding to the maximum current, and outputs a signal about the soft-on time by the current of the current mirror.

4. The switch circuit according to claim 1, further comprising: monitor circuit, which outputs a

signal if the gate voltage is nearly 0 V when the pass transistor is fully turned on.

5. The switch circuit according to claim 1, further comprising: a third monitor circuit, which outputs a signal when the gate-to-source voltage of the pass transistor is less than a predetermined value.
